

- (a) Find the incident power density as per AM1.5G spectrum i.e., integrate the AM1.5G spectrum over the wavelength to estimate the incident power.
- (b) Find the total number of photons in the AM1.5G spectrum
- (c) Find the total number of photons with $E > 1.5\text{eV}$ in AM1.5G spectrum
- (d) Generate a plot between maximum value of J_{sc} and E_g . As discussed in lectures, maximum value of J_{sc} can be obtained by integrating the photons with $E > E_g$ in the AM1.5G spectrum
- (e) For a material with $E_g = 1.5\text{eV}$, $n_i = 10^6\text{cm}^{-3}$, $k_b = 10^{-10}\text{cm}^3/\text{s}$, and $W = 1\text{mm}$, plot the expected J-V characteristics. Assume $J_{dark} = 0$ and consider radiative recombination as done in lectures. Use J_{sc} from part (d)
- (f) Plot the variation of power generated ($J \cdot V$) as a function of V . Estimate the maximum power generated by this solar cell.
- (g) Estimate the power conversion efficiency of this solar cell defined as
- Efficiency = Maximum value of power generated / Incident power
- The value for incident power can be obtained from part (a).

$$\begin{aligned}
 \text{(a) Incident Power} &= \int (\text{Global tilt irradiance}) (\text{change in wavelength}) \\
 &= \int \phi d\lambda \\
 &= \sum \phi \Delta\lambda \text{ for excel calculations}
 \end{aligned}$$

On python,
`np.trapezoid` is used to integrate the data.

The integration value comes out to be 1000.4 W/m^2 .

$$\frac{1 \text{ W}}{\text{m}^2} = \frac{1000 \text{ mW}}{(100)^2 \text{ cm}^2} = 0.1 \frac{\text{mW}}{\text{cm}^2}$$

$$\therefore \text{Incident power} = 100.04 \text{ mW/cm}^2$$

(b) We know that, energy of a photon,

$$E = \frac{hc}{\lambda}$$

For finding no. of photons in the spectrum,

$$\phi_{\text{AMI-5G}} = N \frac{hc}{\lambda}$$

On Excel,

$$(\phi_{\text{AMI-5G}})(\Delta\lambda) = Nhc$$

$$\Rightarrow N = \frac{(\phi_{\text{AMI-5G}})(\Delta\lambda)}{hc} \frac{\text{photons}}{\text{m}^2\text{s}}$$

This N is the number of photons corresponding to that wavelength.

$$N = \frac{\int_{\lambda_{\min}}^{\lambda_{\max}} \phi_{AMI.5G} d\lambda}{hc}$$

$$\Rightarrow \sum N = \frac{\sum_{\lambda=\lambda_{\min}}^{\lambda_{\max}} \phi_{AMI.5G} \Delta\lambda}{hc}$$

∴ After adding all the number of photons corresponding to all the wavelengths, the total number of photons,

$$N_{tot} = 4.3063 \times 10^{21} \frac{\text{photons}}{m^2s}$$

$$(c) \quad E > 1.5 eV$$

$$E = \frac{12400}{\lambda(\text{in } \text{\AA})}$$

$$\Rightarrow \lambda = \frac{12400}{1.5} = 8266.67 \text{ \AA} \text{ or } 826.667 \text{ nm}$$

No. of photons corresponding to $\lambda < 826.667 \text{ nm}$

$$\sum_{\lambda=280}^{\lambda=826} N = \frac{\sum_{\lambda=280}^{\lambda=826} \phi_{\text{AMI-561}} \Delta\lambda}{hc}$$

The total Number of photons come out to be,

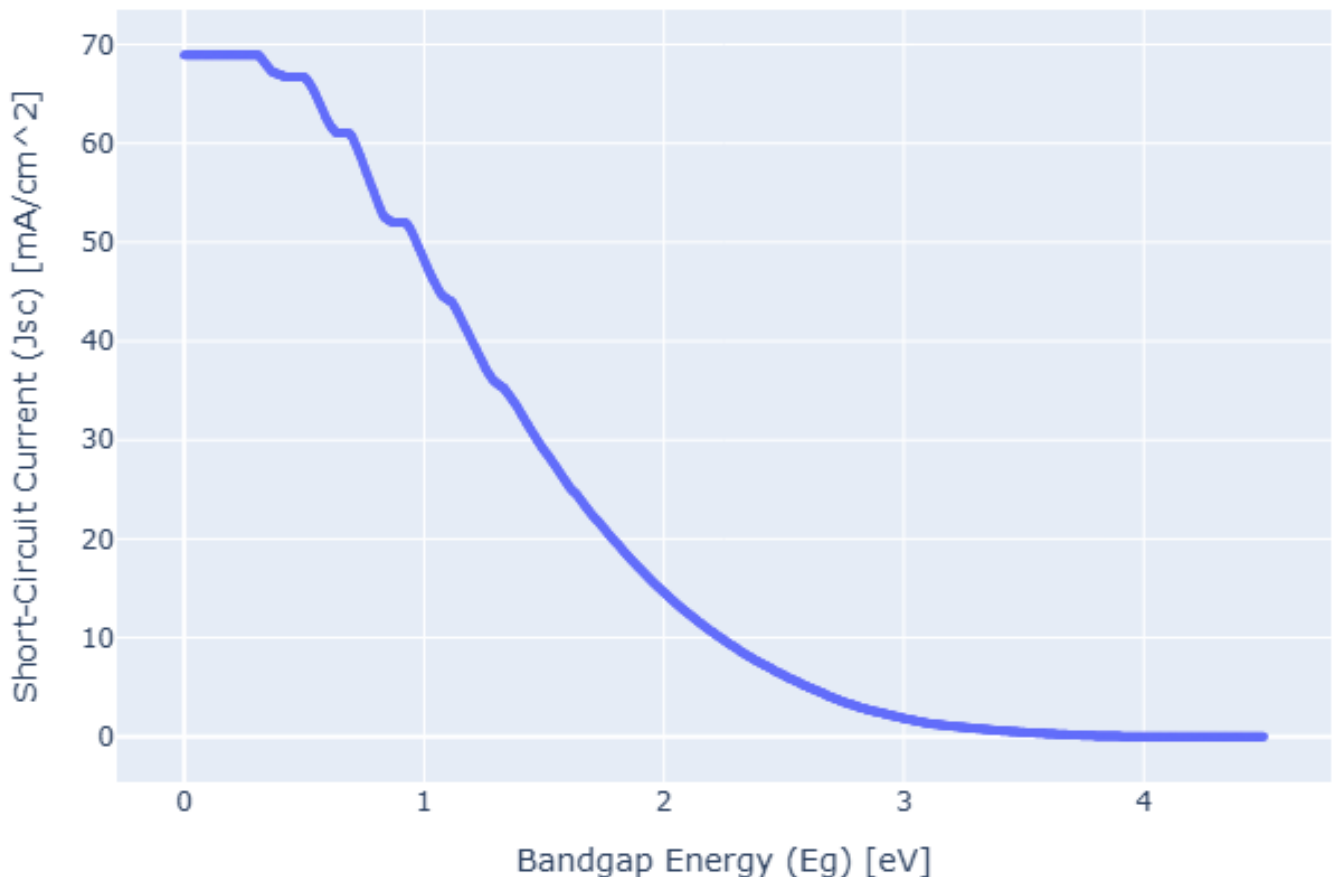
$$N_{>1.5\text{eV}} = 1.81 \times 10^{21} \frac{\text{photons}}{\text{m}^2\text{s}}$$

$$(d) \quad J_{sc}^{\text{max}} = q \int_{\lambda=\lambda_{\min}}^{\lambda=\frac{1240}{E_g}} \frac{\phi_{\text{AMI-561}}}{\frac{hc}{\lambda}} d\lambda$$

On python,
Iterating the calculation done in part (c), and also multiplying with charge on electron, we get,

$$\frac{1 \text{ A}}{\text{m}^2} = 0.1 \frac{\text{mA}}{\text{cm}^2}$$

Maximum Theoretical Short-Circuit Current (J_{sc}) vs. Bandgap Energy (E_g)



$E_g \in \{0, 4.5\}$ with step of 4.5×10^{-3} .

$$\begin{aligned} \text{For } J_{sc}^{\max} @ 1.5 \text{ eV} &= q N_{\text{photon}}^{E > 1.5 \text{ eV}} = 1.8 \times 10^{21} \times 1.6 \times 10^{-19} \\ &= 288.39 \text{ A/m}^2 \\ &\text{or } \underline{\underline{28.84 \text{ mA/cm}^2}} \end{aligned}$$

$$(e) E_g = 1.5 \text{ eV}$$

$$J_{\text{dark}} = 0$$

$$n_i = 10^6 \text{ cm}^{-3}$$

Radiative recombination

$$k_b = 10^{-10} \text{ cm}^3/\text{s}$$

J_{sc} from part (d)

$$W = 1 \mu\text{m}$$

Plot J-V characteristics

$$= 10^{-4} \text{ cm}$$

$$J = \cancel{J_{\text{dark}}}^0 - q \int (G - R) dx$$

$$q \int \cancel{G} dx + q \int R dx$$

J_{sc}

$$\begin{aligned} \int R dx &= \int k_b np dx \\ &= \int k_b n_i^2 e^{qV/kT} dx \\ &= k_b n_i^2 e^{qV/kT} W \end{aligned}$$

$$J = \cancel{-J_{\text{sc}}}^{28.84} + k_b n_i^2 e^{qV/kT} \times W$$

$$J = q \left[- \int_{\lambda = \lambda_{\text{min}}}^{\lambda_{\text{max}} = \frac{1240}{E_g}} \frac{\phi_{\text{AMI-86}}}{\frac{hc}{\lambda}} d\lambda + \underbrace{(10)^{-10} (10^6)^2 e^{\frac{qV}{kT}} (10^{-4})}_{\text{cm}^{-2}\text{s}^{-1}} \right]$$

28.84

$$qk^{-1}T^{-1} = (1.602 \times 10^{-19})(1.38 \times 10^{-23})(300)^{-1}$$

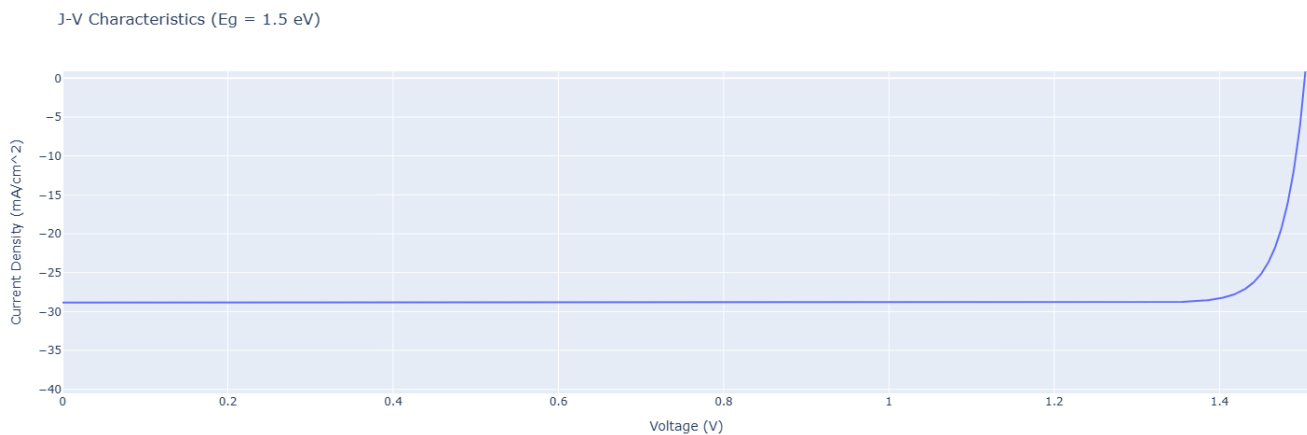
$$= 38.681$$

$$J = \underbrace{-28.84}_{\text{mA/cm}^2} + \underbrace{10^{-2} q e^{38.681V}}_{\text{A/cm}^2}$$

$$J = -28.84 + 10^{-5} q e^{38.681V} \quad \} \text{ mA/cm}^2$$

Done on python,

J_{sc} was used from part (d), and the equation mentioned above was plotted



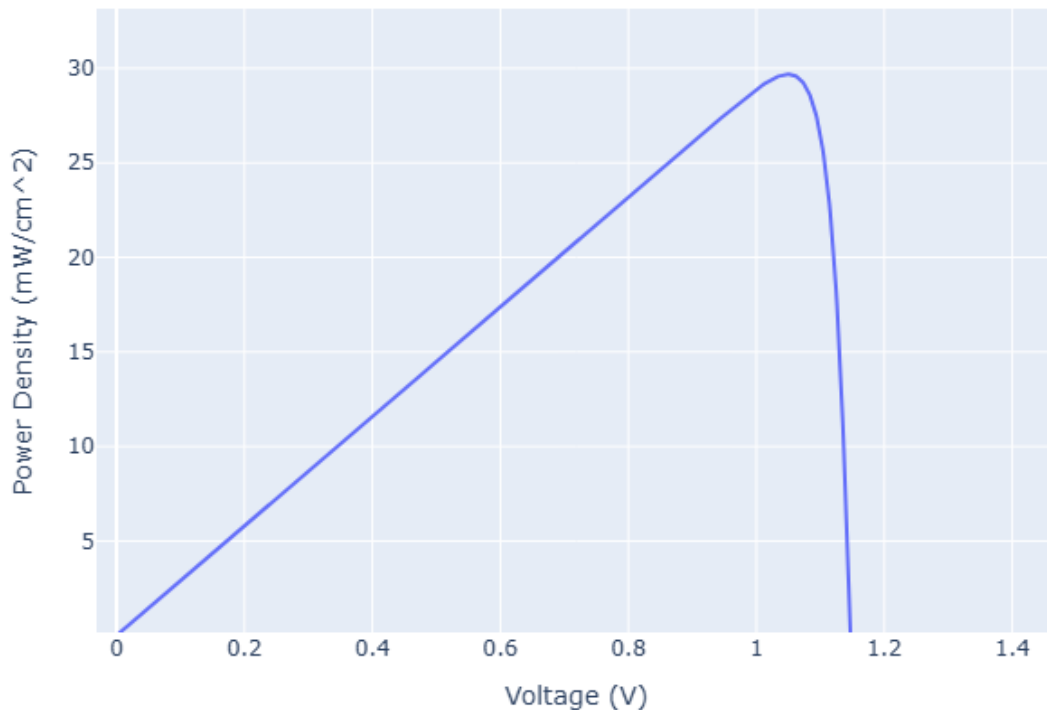
@ $J=0$; $V = 1.5 \text{ V}$

Plotted the values of J for $V \in \{0, 1.5\}$ at step size of 1.5×10^{-3}

$$(f) P = -J \cdot V$$

(-ve sign was used to flip the plot)

Power vs Voltage



Plotted on python,
values of $V \in \{0, 1.5\}$ with a step size of 1.5×10^{-3}

$$(e) P_{\max} = 29.68216 \text{ mW/cm}^2 \text{ @ } V = 1.048048 \text{ V}$$

$$\eta = \frac{P_{\max}}{P_{\text{inc}}} = \frac{29.68216}{100.04} \approx 0.2967 \text{ or } \underline{\underline{29.67\%}}$$